A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 05

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Low liposolubility and low persistence **Also accept** • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1 Bioaccumulation:** build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown **1 Biomagnification:** increase in concentration of the pollutant at successive trophic levels of a food chain **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **bioaccumulation** → a pollutant building up inside one organism over time • **biomagnification** → concentration rising at each step along a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** They are intended to be toxic (to the target species) **1** They are deliberately released / spread into the environment (rather than being an accidental or waste release) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1 Mutagen:** an agent that changes the chemical structure / base sequence of DNA (causes mutations), eg ionising radiation, UV, PCBs, cadmium, asbestos **1 Carcinogen:** a mutagen that causes cancer / uncontrolled cell division producing a tumour **1 Teratogen:** an agent that causes birth abnormalities by preventing normal gene expression (inhibiting proteins / enzymes) without changing DNA structure, eg mercury, 2,4,5-T **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **ionising radiation** → radiation with enough energy to remove electrons and form ions **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Teratogens do not change the DNA structure, so the abnormality is not in the genes and cannot be passed on **1** Mutagens change DNA – if the mutation is in a gamete / egg / sperm (gonadic effect) it can be inherited **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1 Somatic:** mutation in a general body cell, affecting the individual only (eg cancer / tumour) **1 Gonadic:** mutation in cells of the ovaries or testes / gametes / embryo, which may cause abnormalities in offspring **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.4	**1 mark each** 1 mark for each of the 3 points below. **1** Two or more pollutants whose effects interact to produce a different (usually more serious) effect **1** The pollutants themselves do not react to form a new pollutant (that would be a secondary pollutant) **1 Example:** ozone damages leaf cuticles so sulfur dioxide can cause more damage to the exposed cells – smoke and sulfur dioxide in smogs – neonicotinoids with fungicides **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **synergism** → two pollutants together doing more damage than the sum of each alone	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 239-240; AQA spec 3.4.1.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** A waste is assumed to cause pollution until research shows it is unlikely to do so **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	1

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution **1** If polluters can escape their responsibilities / penalties are low, there is little incentive to comply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
03.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Marine discharges where currents dilute and disperse emissions **1** Emissions downwind of urban areas **1** Not discharging waste onto permeable rock above an aquifer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeable** → how easily water passes through / lets water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Net held with its straight edge on the river bed facing the current (open to the flow) **1** The bed in front is disturbed by kicking / scraping a boot for a standard number of kicks and time (eg 30 s) **1** Same substrate type, water depth and flow rate at both sites – same net mesh – repeat samples – invertebrates identified with a key and counted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Difficult to standardise the vigour / area kicked **1** Buried organisms or those fixed to stones are not dislodged **1** Some organisms swim and avoid the net **1** Flow rate varies and affects capture **1** Fragile organisms destroyed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** A fixed area of bed is enclosed by a frame so the sampled area is standardised – side nets reduce escapes **1** But only works in fast enough flow and is hard to set on rough substrates / deep water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Pollutant / oxygen levels fluctuate (eg with flow, time of day) so one reading may miss a pollution event **1** Sensitive taxa (stonefly, mayfly larvae) disappear and tolerant ones (midge larvae, hoglouse) remain, integrating conditions over weeks / months (biotic index) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 395-396, 415-416; AQA spec 3.7.2.1.	
05.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (44 - 8) / 50 = 0.72 m yr^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (m yr^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Abstraction (for irrigation) exceeds the rate of natural recharge by infiltration / percolation of precipitation **1** The dynamic equilibrium between inflow and outflow has been upset so the stored volume declines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Abstraction** → water taken out of a river, lake or aquifer • **infiltration** → water soaking into the soil surface • **percolation** → water moving downwards through soil or rock	2
05.3	**1 mark each** 1 mark for each of the 3 points below. **1 Porosity:** a high proportion of the rock volume is space that can hold water (chalk, limestone, sandstone) **1 Permeability:** the spaces are interconnected and large enough for water to flow through (clay is porous but not permeable) **1** Impermeable rock (granite, clay) beneath to stop water escaping, and permeable rock / soil above to allow recharge **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **Permeability** → how easily water passes through / lets water through • **Impermeable** → does not let water through	3
05.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Salt water incursion:** seawater flows into the aquifer to replace the fresh water, making it unusable for irrigation (osmotic dehydration of crops) **1 Subsidence:** emptied pore spaces no longer support the rock so it compacts, damaging buildings and pipes (eg Mexico City) **1** Drying up of groundwater-fed rivers, lakes and wetlands as the water table falls below their beds **1 Ecological impacts:** plants needing high water availability die – wetland animals lost – food chain effects **1 Reduced supplies:** abstraction may suddenly have to stop after years of hidden decline **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **abstraction** → water taken out of a river, lake or aquifer **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** UV light splits diatomic oxygen (O2) into two monatomic oxygen atoms (O) **1** Monatomic oxygen reacts with diatomic oxygen to form ozone (O3) **1** UV light also splits ozone into O2 and O – the reactions form a dynamic equilibrium which absorbs UV (converting it to chemical energy / heat) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
06.2	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2 Useful:** boiling points near ambient temperature so easily liquefied under low pressure (aerosol propellants, fridge / air-conditioning coolants) – dissolve grease but do not damage electrical components (solvents) – non-flammable – non-toxic **up to 2 Persistence:** chemically stable / unreactive in the troposphere so they remain long enough to be carried up to the stratosphere **up to 2 Dissociation by UV:** in the stratosphere higher UV levels break the carbon-chlorine bonds releasing chlorine free radicals **up to 2** Chlorine reacts with monatomic oxygen (and ozone), preventing ozone formation – chlorine is regenerated so one atom destroys many ozone molecules – bromine / iodine act similarly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **Persistence** → does not break down / stays in the environment for a long time **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	6

Do not write outside the box

Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** UVB energy is absorbed by cells and breaks up biological molecules / DNA causing mutations, skin cancer, cataracts, sunburn (human health) **1** Damage to leaf tissue and reduced photosynthesis / crop yields (plants) **1** Damage to marine algae, phytoplankton, corals and planktonic larvae (marine organisms), reducing productivity and food webs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.4	1 mark for the point below. **1** An area where the total ozone column falls below 220 Dobson Units (normal is about 300 DU) - a level never recorded before CFC pollution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
06.5	1 mark for the point below. **1** Concentration varies with altitude and the stratosphere is not normally sampled – Dobson Units measure the total column thickness instead **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
06.6	1 mark for the point below. **1** The British Antarctic Survey at Halley detected higher UV at ground level, showing ozone above had decreased (satellites above the stratosphere cannot measure downward UV directly) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-109; AQA spec 3.2.1.3.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Its total annual discharge	1

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Public supplies cannot match consumer demand / the water is often cleaner than rivers **1** It also reduces urban flooding after heavy rain – surplus can be used for aquifer recharge **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** The reservoir is created in the estuary so no farmland / settlements need to be flooded **1** But it changes important inter-tidal habitats, obstructs shipping and can be polluted by activities anywhere in the river's catchment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-123; AQA spec 3.2.2.5.	
08.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (52 - 40) / 40 x 100 **1** = 30% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Sulfur dioxide came mainly from burning coal – domestic coal fires were controlled by the Clean Air Acts and coal-fired power stations have closed or been fitted with flue gas desulfurisation **1** Low-sulfur fuels are now used, and sulfur can be removed from the fuel before combustion **1** Nitrogen oxides are produced whenever any fuel is burned in air at high temperature, so they are not removed by changing the fuel – road traffic in the city centre has continued to grow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **flue gas desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone	3
08.3	**1 mark each** 1 mark for each of the 3 points below. **1 Electrostatic precipitator:** the flue gases pass charged wires that give the particulates an electric charge, so they are attracted to oppositely charged collecting plates and can be shaken off and removed **1 Catalytic converter:** a platinum / rhodium catalyst converts oxides of nitrogen, carbon monoxide and unburnt hydrocarbons into nitrogen, carbon dioxide and water **1 Low-NOx burner:** the fuel is burned at a lower temperature, or in stages, so less of the nitrogen and oxygen in the air combines to form oxides of nitrogen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Electrostatic precipitator** → charged plates that attract and collect particulates • **particulates** → fine solid particles / smoke and dust • **Catalytic converter** → a catalyst that converts NOx, CO and unburnt hydrocarbons to nitrogen, carbon dioxide and water	3
08.4	**1 mark each** 1 mark for each of the 3 points below. **1** Ground-level ozone is a secondary pollutant: it is not emitted directly but formed in the air by sunlight acting on oxides of nitrogen and volatile organic compounds **1** Nitrogen dioxide is split by ultraviolet light to give nitrogen monoxide and an oxygen atom, which joins an oxygen molecule to make ozone **1** With less NOx there is less of the raw material for these photochemical reactions, so less ozone and fewer PANs are produced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • ground-level ozone described as protecting us from UV - that is stratospheric ozone; at ground level it is a pollutant	3

Do not write outside the box

Question	Marking guidance	Marks
08.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Lichens differ in their sensitivity to sulfur dioxide, so the species present show the average pollution level over a long period rather than at one instant **1** They need no power supply or expensive equipment and can be surveyed at many sites, giving good spatial coverage cheaply **1** They integrate the effects of pollution over years, so a short pollution episode may not show up **1** They indicate sulfur dioxide well but say little about nitrogen dioxide, ozone or particulates, which are now the city's main problems **1** Other factors affect them too, such as the pH and texture of the bark or stone they grow on, shade and humidity, so a survey can be misleading unless these are standardised **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **indicator species** → a species whose presence or absence shows the conditions present • **particulates** → fine solid particles / smoke and dust **Do not credit** • 'lichens die where the air is polluted' without naming the pollutant - it is their sensitivity to sulfur dioxide that is used	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	
09.1	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Raise river banks / build sea walls, flood barriers or wave screens **1** Permeable urban surfaces (gravel, soil) instead of concrete / tarmac to slow runoff **1** **River flow management:** upstream dams / low soil dams / afforestation / maintained flood plains to delay runoff **1** Raised buildings on stilts or floating houses **1** Pumping rainwater from land below river level **1** Managed retreat - abandoning land where protection costs exceed value (eg Happisburgh) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Permeable** → how easily water passes through / lets water through **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Where the cost of flood prevention is greater than the value of what is protected, or defences would be ineffective **1** Land is abandoned as water levels rise / coast erodes – defences used only where land is very important **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Road surfaces deform as tar melts - need tarmac with a higher melting point **1** Rail track expands and buckles above the temperature it was pre-stretched for (27 °C in UK) - relay with higher pre-stress **1** Bridge damage from high river flows / debris – landslides on deforested slopes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
09.4	**1 mark each** 1 mark for each of the 3 points below. **1** CO2 is captured from the flue gases (or carbon removed from the fuel before combustion) **1** Transported by pipeline / ship / tanker and stored in depleted oil or gas fields, aquifers, or used for secondary oil recovery **1 Limitation:** developmental / expensive, uses energy (reducing efficiency), risk of leakage, limited storage sites **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carbon capture** → capturing carbon dioxide from flue gases and storing it underground	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 99-100, 105-108; AQA spec 3.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Carbon:** combustion of fossil fuels and biomass moving carbon from long-term stores to the atmosphere – deforestation reducing photosynthesis – ploughing / drainage increasing aerobic decomposition – new anaerobic environments (paddies, landfill, reservoirs, livestock) releasing methane – marine pollution reducing phytoplankton – consequences: rising CO2 and CH4, climate change, ocean acidification, reduced soil organic matter • **Nitrogen:** Haber process doubling fixation – legume cultivation – drainage increasing nitrification and reducing denitrification – combustion NOx – sewage and fertiliser leaching – consequences: eutrophication, smogs, acid rain, N2O warming, NOx toxicity • **Counter-strategies carbon:** renewables / nuclear replacing fossil fuels, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, increasing soil organic matter (low tillage, mulching), CCS (pre / post combustion, storage) • **Counter-strategies nitrogen:** reduced combustion, catalytic converters / urea sprays, biological fixation instead of Haber, legumes and crop rotation, management of biological wastes as fertiliser, low-solubility fertilisers, timing, buffer strips, low tillage, nitrate vulnerable zones • **Evaluation:** some strategies well established and cheap (rotation, buffer strips), others expensive or unproven (CCS) – scale of fossil fuel dependence – feedback and time lags – link to circular economy and low-temperature natural processes – judgement	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **nitrification** → ammonium oxidised to nitrite and then nitrate • **denitrification** → nitrate converted back to nitrogen gas • **leaching** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	
	Model answer - Level 3 (8 marks)	
	Humans have shifted the carbon cycle by burning fossil fuels and biomass, moving carbon from long-term stores into the atmosphere faster than photosynthesis and ocean uptake remove it, while deforestation reduces photosynthesis, ploughing and drainage speed aerobic decomposition, and paddies, landfills and livestock create new anaerobic sources of methane. The nitrogen cycle has changed as much: the Haber process has roughly doubled fixation, legume crops add more, drainage increases nitrification and reduces denitrification, combustion produces NOx, and leached fertiliser and sewage nitrate cause eutrophication, while nitrous oxide adds to warming.	
	Carbon can be counteracted by replacing fossil fuels with renewables and nuclear power, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, and increasing soil organic matter through low tillage and mulching, with carbon capture and storage as an unproven option. For nitrogen, catalytic converters and urea sprays reduce combustion NOx, biological fixation by legumes and rotation can replace manufactured nitrate, biological wastes can be managed as fertiliser, and low-solubility fertilisers, careful timing and buffer strips reduce leaching.	
	Some strategies are cheap and established - rotation, buffer strips, fertiliser timing - but others are expensive or untested, and the scale of fossil-fuel dependence, together with feedback and time lags, means the atmosphere will respond slowly. The strategies that mimic natural low-temperature cycling, as in the circular economy, are the most sustainable; the most effective is simply to leave carbon in the ground.	
	Weaker answer - Level 1 (3 marks)	
	Humans have changed the carbon cycle by burning fossil fuels which releases carbon dioxide, and by cutting down trees. They have changed the nitrogen cycle by using fertilisers which run off into rivers and cause eutrophication. To counteract this we could use renewable energy instead of fossil fuels and plant more trees. We could use less fertiliser. These would help the cycles go back to normal.	
	What makes the difference	
	• Level 1 names one or two changes per cycle; Level 3 covers several processes for each (decomposition, methane sources, Haber process, nitrification, NOx).	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 connects each counter-strategy to the process it reverses; Level 1's strategies are general.	
	• The evaluation in Level 3 distinguishes established from unproven strategies and comments on scale and time lags - Level 1 has no evaluation.	
10.2	1 mark for the point below. **1** The average time an atom / molecule remains in the reservoir before being transferred out (volume / transfer rate) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-157; AQA spec 3.2.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • **Fossil fuels:** choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports – new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • **Nuclear:** high energy density reduces transport – waste properties (half-life) determine storage – thorium and fusion prospects • **Renewables:** matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting) – intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity – low-impact designs (helical turbines, VAWT, tidal lagoons) • **Energy conservation reduces demand:** transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use	25
	– life-cycle / embodied energy • **Evaluation:** no resource is impact-free – knowledge allows the best mix and mitigation – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

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Question	Marking guidance	Marks
11.2	**Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Composition and structure:** N_2, O_2, Ar, CO_2, ozone – troposphere and stratosphere – pressure and liquid water – gases for biological molecules and industry • **Energy processes:** UV absorbed by O / O_2 / O_3 dynamic equilibrium – visible light absorbed at the surface – infrared absorbed by greenhouse gases (33 °C warming) – heat distribution by winds and ocean currents – transport of water vapour – energy budget and albedo • **Global climate change:** anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species – difficulties of monitoring and prediction – control strategies and agreements • **Ozone depletion:** CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • **Other threats:** aerosols, smoke, land-use change – interconnection of processes	25
	• **Evaluation:** which functions are most at risk – success of ozone controls vs slow progress on greenhouse gases – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	

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